

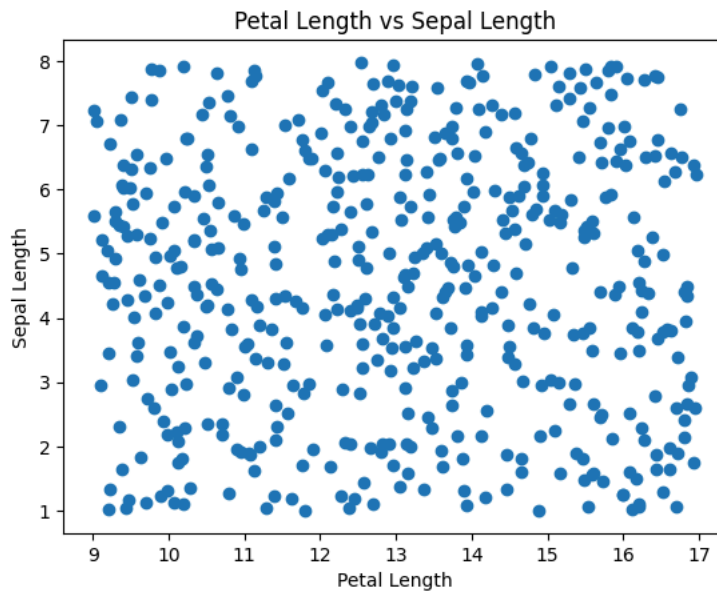
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
import numpy as np

# Generate a random dataset
num_rows = 500 # Using the count from the original data.describe()
data = pd.DataFrame({
    'sepal_length': np.random.uniform(1.0, 8.0, num_rows),
    'sepal_width': np.random.uniform(2.0, 15.5, num_rows),
    'petal_length': np.random.uniform(9.0, 17.0, num_rows),
    'petal_width': np.random.uniform(0.3, 12.5, num_rows)
})
print(data.head())
```

	sepal_length	sepal_width	petal_length	petal_width
0	3.311134	5.342604	11.311591	10.060833
1	7.634538	10.099390	12.703317	12.129733
2	6.075674	2.623623	14.944248	3.989570
3	7.978072	11.855360	12.544861	4.737145
4	3.253332	14.399786	10.130847	8.427228

```
plt.scatter(data["petal_length"], data["sepal_length"])
plt.title("Petal Length vs Sepal Length")
plt.xlabel("Petal Length")
plt.ylabel("Sepal Length")
plt.show()
```



```
plt.hist(data["sepal_length"], bins=15, color="skyblue", edgecolor="black")
plt.title("Distribution of Sepal Length Scores")
plt.xlabel("Score")
plt.ylabel("Frequency")
plt.show()
```

Distribution of Sepal Length Scores



```
print(data.describe())
numeric_data = data.select_dtypes(include=['number'])
print(numeric_data.corr())
sns.heatmap(numeric_data.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Between Variables")
plt.show()
```

	sepal_length	sepal_width	petal_length	petal_width
count	500.000000	500.000000	500.000000	500.000000
mean	4.548244	8.523252	13.011747	6.382410
std	1.990837	3.350053	2.284877	3.490252
min	1.004048	2.002028	9.007251	0.336387
25%	2.938879	5.251953	11.030323	3.327389
50%	4.555662	8.253639	13.049359	6.359187
75%	6.231435	11.075229	14.940998	9.256414
max	7.978072	15.453995	16.957908	12.452079

	sepal_length	sepal_width	petal_length	petal_width
sepal_length	1.000000	0.014145	0.003779	0.043791
sepal_width	0.014145	1.000000	-0.050985	0.011315
petal_length	0.003779	-0.050985	1.000000	0.035264
petal_width	0.043791	0.011315	0.035264	1.000000

Correlation Between Variables

